

LLM-Driven Ontology Learning: From Term Extraction to Upper-Level Categorization for Building Information System Models

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Abstract: Knowledge-intensive Information Systems support high-value corporate tasks. However, building knowledge models requires manual selection and semantic clarification of terminology — a costly bottleneck in enterprise ontology engineering. This work explores automating these foundational steps to reduce manual effort during conceptual modeling. We evaluate the capacity of general-purpose Large Language Models (LLMs) to extract terminology and classify it into upper-level meta-types without specialized fine-tuning. We introduce an empirical framework comparing four term-extraction strategies: LLM + Corpus, LLM Generated, Named Entity Recognition (NER) + Corpus, and TF-IDF, applied to 40 Brazilian Pre-Salt scientific articles. To standardize classification, we leverage LLMs to generate Natural Language Definitions (NLDs) for the extracted terms, explicitly using these NLDs as semantic pivots to improve categorization into a multi-layered hierarchy integrating Basic Formal Ontology (BFO), GeoCore, and GeoReservoir ontologies. The resulting artifacts underwent blind validation by domain experts. Results demonstrate that non-specialized LLMs produce higher semantic quality in term extraction and taxonomic categorization compared to traditional statistical methods. The LLM + Corpus pipeline proved the most robust strategy. Ultimately, this work presents a validated paradigm for modernizing conceptual modeling and bootstrapping domain taxonomies through strategic enterprise-scale automation during conceptual modeling.

1 INTRODUCTION

The design of knowledge-intensive Information Systems depends critically on the quality of their underlying conceptual models. Ontologies, as formal and explicit specifications of a shared conceptualization (Studer et al., 1998), provide the structural backbone for these systems. They enhance semantic clarity, thereby eliminating ambiguities across data models and software applications. Consequently, ontology engineering has become increasingly vital in corporate environments.

In this work, we focus on the foundational step of domain ontology construction: acquiring and structuring a specialized vocabulary. This vocabulary is designed to serve as the core structural backbone for an industrial software application that retrieves and analyzes analogous deposits in petroleum pre-salt exploration. Based on this conceptual framework, reservoir information from diverse heterogeneous sources is stored in a uniform database for similarity analysis.

Despite recent advances, Ontology Engineering still faces significant bottlenecks, particularly during the knowledge acquisition phase. Terminology elicitation and the semantic clarification of vocabulary require the involvement of highly specialized human resources for extended labor sessions. To ensure semantic interoperability, these vocabularies must be mapped to hierarchical structures (Guarino, 1998), spanning from *top-level* to *domain* ontologies. Populating this robust framework manually cannot keep pace with the exponential growth of unstructured data. We aim to automate the steps of terminology selection and classification, or at least significantly reduce human intervention to build the artifact.

While Large Language Models (LLMs) offer a potential solution to accelerate this process, approaches that rely on domain-specific fine-tuning (Sousa et al., 2023), (Lo et al., 2024) often require the curation of costly, manually annotated datasets. This work investigates a more accessible path: applying a general-purpose foundation model without domain-specific

fine-tuning. We raise a critical research question: *Can a non-specialist LLM produce semantically precise artifacts, such as term extraction and taxonomic categorization into formal hierarchies, to bootstrap ontology construction?*

We design and compare four distinct pipelines for term extraction: (1) Corpus-Extractive LLM, (2) LLM Generated, (3) Named Entity Recognition (NER) Model + Corpus, and (4) a traditional TF-IDF baseline. We then pivot on the LLM’s capacity to generate Natural Language Definitions (NLDs), which are concise, descriptive texts that clarify a term’s meaning, to improve the automated classification process (Lopes Junior, 2025). The framework is applied to the complex domain of Brazilian Pre-Salt petroleum geology using a curated corpus. The results undergo blind empirical validation by domain experts, yielding a preliminary conceptual taxonomy composed of relevant, correctly categorized terms that serves as the basis for future axiomatization and ontology construction.

The remainder of this work is organized as follows: Section 2 reviews foundational literature; Section 3 details our comparative empirical approach; Section 4 describes the four pipelines, NLD generation, categorization into upper-level ontologies, and blind validation steps; Section 5 presents the empirical findings and discussion; and Section 6 concludes the study, summarizing insights and next steps.

2 BACKGROUND

This section reviews the current state of the art in LLM applications for Ontology Learning.

2.1 Ontology Learning and Machine Learning

Ontology Engineering’s primary bottleneck remains knowledge acquisition. This has led to the sub-field of Ontology Learning, which seeks to automate the extraction of concepts, relations, and axioms from unstructured text (Khadir et al., 2021). Our work focuses on the foundational upstream tasks: term extraction and hierarchical categorization.

Traditional automated approaches have relied on statistical methods, using frequency-based analyses (e.g., TF-IDF) to extract terms from a domain corpus Garcia et al. (2020). However, these methods often fail semantically, unable to separate critical domain concepts from noise. Contextual embeddings, such as BERT (Devlin et al., 2019), enhanced semantic capability for downstream tasks like alignment and

subsumption prediction (Chen et al., 2023), (He et al., 2022), (Sousa et al., 2023). Nevertheless, these specialized models still require costly domain-specific fine-tuning and labeled data.

2.2 Ontology Learning and Large Language Models

Systematic reviews confirm the significant impact of LLMs in reducing reliance on human domain experts throughout the ontology engineering lifecycle (Amalki et al., 2025), (Kommineni et al., 2024). While pre-trained LLMs have historically struggled with zero-shot term typing and ontology construction (Babaei Giglou et al., 2023), (Giglou et al., 2024), fine-tuning approaches have emerged as a relevant strategy (Lo et al., 2024), though they remain data-intensive.

Advanced LLMs have demonstrated the capacity to generate OWL ontology drafts directly from requirements, outperforming novice human engineers but still requiring oversight (Saedizade and Blomqvist, 2024), (Lippolis et al., 2025), (Fathallah et al., 2024), (Bakker et al., 2024). Crucially, recent work has demonstrated the high performance of language models in classifying domain entities into top-level ontology concepts using NLDs (Lopes Junior, 2025). This highlights that NLDs are a highly effective textual representation for classification. Our work builds upon this insight by empirically comparing extraction methods and leveraging NLDs as the foundational pivot to map raw terms into a formal upper-level hierarchy.

3 A NEW APPROACH FOR TERM EXTRACTION AND CLASSIFICATION

Following (Lopes Junior, 2025), we empirically compare four term-extraction pipelines: (1) LLM + Corpus, (2) LLM Generated, (3) NER + Corpus, and (4) TF-IDF. Extracted terms feed into a unified classification module where an LLM generates a Natural Language Definition (NLD) for each. Using these NLDs as semantic pivots, terms are categorized into a formal hierarchy integrating BFO (Arp et al., 2015), GeoCore (Garcia et al., 2020a), and GeoReservoir (Ciconeto et al., 2022), ensuring interoperability with established geological standards.

Finally, we execute a blind, multi-expert empirical validation to quantitatively and qualitatively evaluate the resulting conceptual artifacts (Terms, NLDs, and

Categories).

4 METHODOLOGY

We have organized the methodology of this work into four phases. The first phase is preparing the curated corpus. The second phase consists of term acquisition (performed using four distinct pipelines). The third phase involves generating natural language definitions (NLDs) and classifying the terms. Lastly, the fourth phase consists of a comparative assessment of the four pipelines across three critical dimensions: term extraction coverage, NLD accuracy, and taxonomic categorization.

4.1 Phase 1: Corpus Preparation

The foundation of our experiment is a highly specialized corpus comprising 40 curated scientific articles on the Brazilian Pre-Salt. A senior domain specialist rigorously selected these articles to ensure the corpus reflects the most optimized and authoritative knowledge available. This focused approach ensures the corpus is peer-reviewed, representative, and authoritative for the subsequent term extraction and classification tasks, providing the necessary informational density while aligning with the limited human resources.

The method automatically extracted text from all documents using a simple Python script that leveraged the open-source library `pdftotext`. We decided to retain the entire extracted text, including abstracts and references, because these sections often contain high-value terms and concept definitions critical for ontology construction.

4.2 Phase 2: Competing Term Acquisition Pipelines

All LLM-based steps used the general-purpose, non-specialized Gemini 2.5 Pro foundation model throughout the process. This choice was justified by both its advanced reasoning capabilities, which are crucial for categorization tasks, and its high availability, ease of use, and streamlined API access, which ensure robust, reproducible, and operationally efficient execution. Unless otherwise specified (i.e., Pipeline 3), we have maintained a temperature of 0 for all LLM steps to maximize output determinism.

- **Pipeline 1 (Corpus-Extractive LLM):** This pipeline uses the LLM in a purely extractive mode. We prompted the model to read each of

the 40 article texts and extract all relevant geological terms. The method aggregated the extracted terms through stemming before proceeding to the subsequent classification workflow.

- **Pipeline 2 (LLM Generated):** Pipeline 2 operates zero-shot, prompting the LLM to generate domain terms from its pre-trained weights. A temperature of 0.2 was used to allow slight lexical variance while maintaining factual grounding.
- **Pipeline 3 (Hybrid NER-LLM):** This pipeline tests a hybrid, cross-lingual approach. The specialized multilingual NER model XLM-RoBERTa Large (Moreira et al., 2025)) was allowed to extract terms and then fine-tuned on the PetroGeoNER petroleum-specialized corpus. Then, the method processed the terms and their respective NER labels, which were extracted by this model.
- **Pipeline 4 (Statistical Baseline):** This pipeline replicates the traditional methodology (Garcia et al., 2020b) and serves as our non-AI baseline. It uses the TF-IDF (Term Frequency-Inverse Document Frequency) weighting strategy, one of the most popular and effective methods in information retrieval. Our method ran on the 40-article corpus.

The raw outputs from all four pipelines showed significant variability. To ensure systematic and reproducible aggregation, a normalization and consolidation step was implemented. This process involved applying an English stemmer to normalize terms to their root forms, followed by mapping each stem back to its most frequent lexical form for human readability. The steps produced four distinct, consolidated, and frequency-filtered lists, establishing the robust empirical basis for the subsequent NLD-centric classification workflow.

The results from all distinct pipelines, along with the source code and the specific LLM prompts used for each step, are made available on <https://github.com/BDI-UFRGS/PreSaltOntology>

4.3 Phase 3: NLD-Centric Classification Workflow

This phase implements a single, standardized workflow to process the candidate terms from all four pipelines. We designed this workflow to explicitly test the findings of (Lopes Junior, 2025) by using NLDs as the pivot for classification.

All unique candidate terms from the four consolidated lists were fed into an NLD generation module,

leveraging the LLM to generate a canonical definition. We prompt the LLM with a system instruction defining its role as a senior geoscientist and ontology engineer. For each term, the LLM receives a prompt template instructing it to generate a concise, technically clear, Aristotelian-style definition (e.g., "X is a Y that Z") aligning with (Lopes Junior, 2025) findings.

Receiving one request per term, a Chain-of-Thought (CoT) prompt instructed the model to analyze the term and its corresponding NLD, and then classify it into the most specific category possible. This classification strictly followed a fallback hierarchy: first, GeoReservoir (a more specialized ontology), then GeoCore (a core ontology), and finally BFO (the top-level ontology).

4.4 Phase 4: Specialist Evaluation

To evaluate the performance of each pipeline, we designed a blind, multi-expert empirical validation performed by geologists specialized in the geological Pre-Salt domain.

- **Expected terms:** Before any sequential steps, five domain specialists provided a reference set of 100 terms (20 terms each) representing essential concepts for the Pre-Salt domain. This group consisted of highly experienced professionals with advanced academic degrees (PhD and Master's) and extensive combined experience spanning both petroleum industry operations and petroleum academia.
- **Sampling:** The method described in Section 4 created a final sample from each pipeline by selecting the 100 most relevant terms, along with their generated NLDs and assigned categories, from each of the four pipelines, based on its respective relevance ranking mechanism. These 400 entries were combined, anonymized, and randomized into a single list. This fixed limit of 100 terms per pipeline was a pragmatic decision driven by limited human resources and the recognized labor-intensive nature of expert validation, ensuring the empirical assessment remains feasible, rigorous, and non-exhaustive for the domain specialists.
- **Quantitative Metrics:** The key metrics collected for this study were:
 - **Term Relevance:** To measure the quality of the term acquisition (Relevant, Irrelevant, Invalid, Unknown). The same five domain specialists that provided the Expected terms participated in this evaluation.

- **Definition Accuracy:** To measure the quality of the NLD-generation step (Correct, Partially Correct, Incorrect, Ambiguous / Too Vague, Wrong Context, Unknown). Three of the initial five experts participated in this validation, bringing advanced degrees (MSc and PhD) and up to 20 years of combined industry and academic Pre-Salt experience.
- **Category Accuracy:** To measure the quality of the final classification, using a metric to capture the accuracy and specificity (e.g., Correct and Specific, Correct but Unspecific, Incorrect, Unknown). The same three specialists from the previous step participated in this evaluation.

This mixed-methods assessment enables an empirical comparison of the results from each pipeline.

5 RESULTS AND DISCUSSION

This section presents the comparative evaluation of the four ontology learning pipelines: LLM + Corpus, LLM Generated, NER + Corpus, and TF-IDF. Based on the assessment by domain specialists, we analyze performance across four critical dimensions: term extraction coverage, terminological relevance, NLD accuracy, and taxonomic categorization accuracy.

5.1 Term Extraction Coverage

The terms extracted by the pipelines were compared against the expert-defined reference terms (20 terms provided by each of the five specialists) deemed essential for the domain coverage. The LLM + Corpus pipeline achieved the highest coverage, followed by LLM-generated, while NER + Corpus and TF-IDF demonstrated significantly lower recall (Table 1).

A clear distinction emerged in entity type performance. While concrete geological objects (e.g. 'Calcite', 'Coquina') were widely detected by LLM and NER pipelines, a systematic gap existed in the extraction of petrophysical properties and attributes (e.g., 'grain roundness'). This absence was attributed to strict prompt constraints that prioritize "conceptual classes" and exclude numerical values, confirming model adherence rather than capability limitations. Future iterations should explicitly add properties to mitigate this.

A "granularity gap" was also noted, where automated extraction favored generic terms over the specific compound terms that experts had expected. Notably, TF-IDF exhibited the lowest efficacy, failing to capture highly relevant low-frequency terms (e.g.,

Table 1: Term Extraction Coverage and Relevance Distribution per Pipeline

Pipeline	Recall (%)	Relevant (%)	Irrelevant (%)	Unknown (%)	Invalid (%)
LLM + Corpus	60	97.8	2.0	0.2	0.0
LLM Generated	40	96.2	3.8	0.0	0.0
NER + Corpus	32	86.6	11.8	0.8	0.8
TF-IDF	25	46.0	46.4	1.4	6.2

‘Stevensite’). In contrast, semantic LLM methods successfully retrieved these terms.

In summary, the LLM + Corpus pipeline proved the most robust for conceptual entities, offering the best balance between precision and abstraction. While NER + Corpus showed resilience in detecting named entities filtered out by LLM prompt constraints, it lacked semantic depth.

5.2 Term Relevance Evaluation

Table 1 reveals a distinct dichotomy between semantic (LLM-based) and statistical/syntactic (TF-IDF/NER + Corpus) extraction methods. The LLM + Corpus pipeline achieved the highest overall and consensus relevance. This result contrasts sharply with the TF-IDF baseline, which had introduced significant noise (46.4% Irrelevant, 6.2% Invalid). Furthermore, the analysis of evaluator consensus in Table 2 highlights a qualitative advantage of LLMs. While the NER + Corpus pipeline achieved a respectable relevance rate, it generated a high volume of ‘Mixed Feelings’ - this category signifies terms where evaluators lacked consensus, being deemed simultaneously relevant and irrelevant by different specialists.

Finally, a subtle but important distinction emerged between the LLM approaches. While both methods achieved high relevance, the LLM + Corpus pipeline proved superior by leveraging source text as a semantic anchor, yielding slightly higher consensus. It helped mitigate the “domain blindness” of the LLM-generated approach, demonstrating that corpus grounding can help capture the specialized vocabulary of the Pre-Salt domain.

5.3 Natural Language Definition Evaluation

Given the identical, context-devoid definition prompt across all pipelines, performance divergence stemmed solely from the nature of the input terms. The LLM-Generated pipeline achieved the highest accuracy in Table 2. This was attributed to the model’s “internal coherence”, in which it prioritized canonical, theoretical concepts (e.g., ‘Dolomite’) for which it possesses

robust internal definitions.

The LLM + Corpus pipeline had a high rate of ‘Partially Correct’ definitions. This occurred because complex, empirically derived terms lacked the source context needed for the general LLM to generate precise definitions, resulting in generic outputs that were insufficiently nuanced for experts. This exposes the critical methodological limitation of context blindness during the definition phase. These findings demonstrate that defining empirically extracted terms is significantly more demanding than defining theoretical ones. Expert feedback also highlighted a scientific currency limitation: definitions sometimes reflected outdated paradigms (e.g., homogeneous microbial origin for the Pre-Salt carbonates), due to the LLM’s training data’s temporal constraints. Future iterations must incorporate Retrieval-Augmented Generation (RAG) to address both issues by providing necessary context and an updated stream of high-quality domain literature to prevent the propagation of an obsolete paradigm.

5.4 Taxonomic Categorization Evaluation

Taxonomic categorization evaluation results are shown in Table 3. It reveals a fundamental trade-off between terminological stability and domain validity driven by the origin of the input terms. The LLM-Generated pipeline achieved the highest Strict Consensus. The model prioritized canonical, theoretical concepts (e.g., ‘Dolomite’) from its pre-training that elicit unanimous expert agreement. Conversely, the LLM + Corpus pipeline demonstrated superior overall accuracy, achieving the highest Majority Consensus with the lowest rejection rate. While real-world terms extracted from reports often exhibit complexity that sparks taxonomic debate, they represent a more accurate reflection of the Pre-Salt domain than the theoretically “clean” of the generative approach, which saw a higher rejection rate.

The significant proportion of terms achieving Not Strict Consensus rather than Strict Consensus (Correct and Specific) in the LLM + Corpus pipeline (Table 3) exposes a critical methodological bottle-

Table 2: Term Relevance Consensus and NLD Evaluation Metrics

	LLM + Corpus	LLM Generated	NER + Corpus	TF-IDF
Term Relevance Consensus (Count/100)				
Strictly Relevant	91	88	66	25
Strictly Irrelevant	0	0	1	22
Mixed Feelings	9	12	33	53
Natural Language Definition Accuracy (%)				
Correct	53.7	66.0	57.0	34.7
Partially Correct	34.0	24.3	22.3	22.0
Incorrect	9.7	6.7	15.3	9.0
Ambiguous or Too Vague	1.0	2.3	2.0	14.7
Unknown	1.0	0.3	2.0	15.7
Wrong Context	0.7	0.3	1.3	4.0

neck: context blindness. This suggests the LLM’s full reasoning potential is currently constrained by the data architecture. This may also indicate that implementing RAG in the NLD generation phase can improve categorization and elevate majority-consensus instances to strict consensus. Additionally, the limitations of non-semantic methods were evident. NER + Corpus showed high disagreement (45% Majority Correct vs. 23% Strict Correct and Specific), and TF-IDF yielded the highest rate of ”Unknown” results.

The high acceptance in the Table 3 rate further confirms that using the corpus acts as a decisive ’Validity Filter’ for the extracted terms. When accounting for partial success, the LLM + Corpus pipeline achieved a total acceptable categorization rate (Majority Correct and Majority Partially Correct) underscoring its efficacy as the primary knowledge acquisition tool when compared to LLM-generated categorized terms.

5.5 Resulting Conceptual Artifact

The experiment produced a tangible output: a set of validated BFO-based conceptual artifacts representing a preliminary domain taxonomy for Brazilian Pre-Salt petroleum reservoirs (a fragment of which is illustrated in Figure 1). The final consolidation yielded a core set of domain concepts that achieved unanimous consensus among the three expert evaluators regarding both Term Relevance and Category Accuracy.

These terms were formally instantiated into our multi-layered hierarchical structure. For example, extracted geological objects were successfully mapped down the hierarchy (e.g., *BFO:Entity* → *BFO:Continuant* → *BFO:Independent Continuant* → *BFO:Material Entity* → *GeoCore:Earth Material* → *GeoCore:Amount Of Rock* → *Geo-*

Reservoir:Sedimentary Rock → **Shale**), while extracted processes were appropriately anchored (e.g., *BFO:Entity* → *BFO:Occurrent* → *BFO:Process* → *GeoCore:Geological Process* → **Diagenesis**).

Although this generated taxonomy is not fully axiomatized and does not yet encompass complex dependent entities like properties or relationships, it remains a highly significant artifact. It demonstrates that a semi-automated pipeline, driven by NLD generation and expert-in-the-loop validation, can successfully bridge the gap between raw text and formal upper-level structures. It moves beyond simple term extraction to produce semantically grounded, hierarchically consistent artifacts, thereby significantly easing the most labor-intensive initial phases of ontology construction.

The complete validated taxonomic hierarchy and all supplementary materials are available at <https://github.com/BDI-UFRGS/PreSaltOntology>.

6 CONCLUSION

This study evaluated a methodological framework for partially automating knowledge acquisition in domain ontology engineering. Comparing four pipelines, the LLM + Corpus approach proved most robust for term extraction, significantly outperforming TF-IDF and NER. Furthermore, the novel application of NLDs as a semantic pivot proved effective, enabling the automated mapping of raw terms into a formal, multi-layered conceptual hierarchy.

Beyond methodological insights, this process yielded a tangible, expert-validated conceptual artifact tailored to the Brazilian Pre-Salt domain. This domain taxonomy serves as the structural foundation for an enterprise software application designed to re-

Table 3: Consolidated Category Accuracy: Distribution and Consensus Metrics

	LLM + Corpus	LLM Generated	NER + Corpus	TF-IDF
Overall Accuracy Distribution (%)				
Correct & Specific	67.00	64.67	61.67	40.67
Correct but Unspecific	15.00	11.00	20.67	10.00
Incorrect	16.33	22.67	14.33	25.33
Unknown / Cannot Judge	1.67	1.67	3.33	24.00
Consensus Breakdown (Count/100 Terms)				
<i>Strict Consensus</i>				
Correct and Specific	39	43	23	17
Correct but Unspecific	1	1	0	0
Incorrect	3	9	3	4
<i>Majority Consensus (Not Strict)</i>				
Correct and Specific	32	23	45	19
Correct but Unspecific	10	8	10	5
Incorrect	8	14	6	17
Unknown	0	0	0	16
<i>Ambiguity (No consensus)</i>	7	2	13	22
Final Majority Metrics (%)				
Majority Correct	71	66	68	36
Majority Partially Correct	11	9	10	5
Majority Disputed / Rejected	18	25	22	59

trieve and analyze analogous sedimentary deposits analogous to real petroleum reservoirs. Future steps will focus on formal axiomatization to transition this validated hierarchy into a fully realized computational ontology.

Finally, empirical validation revealed a clear trade-off: while the LLM-Generated pipeline provided better definitions due to internal coherence, it yielded less relevant terms. Conversely, the LLM + Corpus approach extracted highly relevant terms but struggled with ambiguity and outdated paradigms. To overcome this context blindness, we intend to integrate RAG into the NLD generation phase in future development of this work. RAG will provide necessary source grounding to resolve ambiguity and ensure scientific currency, paving the way for a scalable, semi-automated knowledge engineering pipeline.

While this article presents an overview of the results, we have also made all supplementary material available, including the code, the curated corpus, the full list of categorized terms, and the corresponding expert evaluations, at <https://github.com/BDI-UFRGS/PreSaltOntology>.

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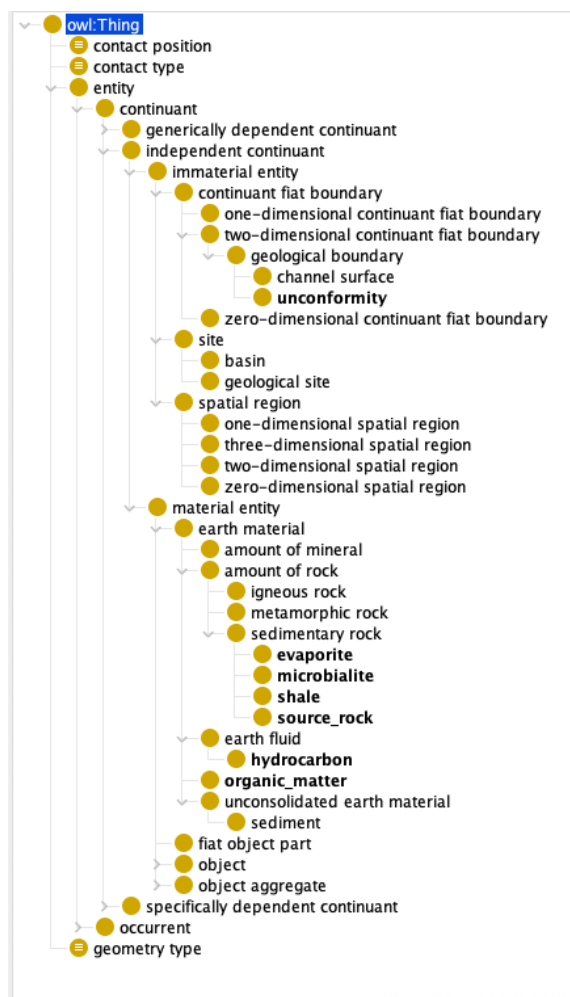


Figure 1: Excerpt of conceptual artifact, containing the validated terms classified into the three hierarchical levels: BFO, GeoCore, and GeoReservoir. Classes in bold indicate the extracted terms.

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